

Dual-Functional Metallized Basalt Fiber with Exceptional Electromagnetic Interference Shielding and Joule Heating Capabilities under Simulated Lunar Temperature Conditions

Che Zhou, Lin Zhao, Lei Hui, Zeyu Yang, Tao Fu, Jian Huang, and Dongzhen Chen*



Cite This: *ACS Appl. Mater. Interfaces* 2026, 18, 39093–39105



Read Online

ACCESS |



Metrics & More



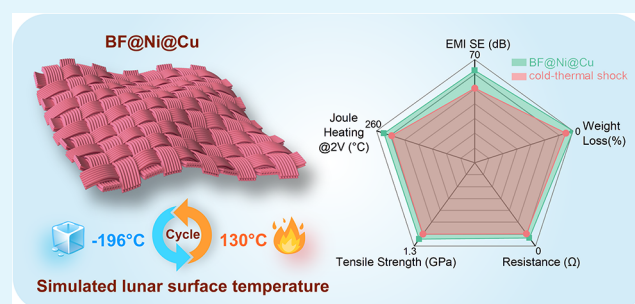
Article Recommendations



Supporting Information

ABSTRACT: The lunar mare region is rich in basaltic minerals, and in situ resource utilization (ISRU) is a fundamental strategy for sustainable extraterrestrial construction. However, combining electroless plating with basalt fibers for such construction leads to a significant mismatch of coefficient of thermal expansion (CTE) between the fiber substrate and metal coating. Under extreme temperature alternations, this mismatch induces interfacial thermal stress concentration, causing coating peeling and performance failure. To address this issue, this study, using commercial terrestrial basalt fiber as an analogue for lunar basaltic materials, proposes an ISRU-inspired metallized fiber composite suitable for wide-temperature-range applications. By sequential electroless nickel plating and copper electroplating on basalt fibers, a nickel–copper-coated basalt fiber fabric (BF@Ni@Cu) was successfully fabricated, exhibiting high electrical conductivity, excellent electromagnetic interference shielding effectiveness (62.59 dB), and significant joule heating performance. The Ni interlayer forms a CTE gradient transition between the basalt substrate and the outer Cu layer, mitigating interfacial thermal stress. After annealing and PDMS encapsulation, the surface reflection characteristics are effectively regulated. To verify reliability under lunar diurnal temperature variations, cold–thermal shock cycle tests simulating the lunar range (from -196 to 130 °C) are conducted. After 30 cycles, the material maintained structural integrity without cracking or peeling, successfully overcoming interfacial thermal stress concentration. Consequently, the EMI shielding and joule heating performance showed only slight degradation, demonstrating excellent temperature shock resistance. This study not only provides a fiber metallization strategy that retains high performance under extreme temperature alternations but also offers a potential technical pathway inspired by ISRU for multifunctional protection and thermal management materials in future lunar base construction, through the design concept of thermal stress alleviation and failure-mode control via a gradient interlayer.

KEYWORDS: in-situ utilization of lunar resource, basalt fiber, surface metallization of fiber, electromagnetic interference shielding, extreme temperature alternations



1. INTRODUCTION

As global lunar exploration activities enter a new phase centered on long-term habitation and resource utilization, the construction of permanent lunar bases capable of withstanding extreme environments has become the next milestone in space exploration.^{1,2} Distinctly different from terrestrial construction, building materials for lunar bases face ultimate challenges from the celestial environment: intense cosmic radiation and solar particle events³ demand that structures possess exceptional electromagnetic shielding capability to protect precision instruments and astronaut health; the alternation of day and night, each lasting approximately 14 Earth days, results in drastic lunar surface temperature fluctuations between -180 and 130 °C,⁴ imposing dual requirements on the thermo-mechanical stability and active thermal management of materials. Therefore, developing novel composite materials that integrate structural load-bearing, electromagnetic protec-

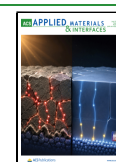
tion, and built-in heating functions is key to overcoming the technological bottlenecks in lunar base construction. Recent advances have demonstrated various flexible composites combining EMI shielding and Joule heating, such as polyimide/carbon nanotube films,⁵ silver-decorated polyimide nanofibers,⁶ Ag nanoparticle-modified cotton fabrics,⁷ MXene/CNT/TPU nonwovens,⁸ and gradient-structured foam-coated fabrics.⁹ These studies have made important contributions to terrestrial applications. However, the exploration of similar

Received: April 20, 2026

Revised: June 18, 2026

Accepted: June 24, 2026

Published: June 29, 2026



multifunctional composites designed for the coupled effects of extreme thermal cycling and intense radiation on the Moon especially within the context of in situ resource utilization remains largely open.^{10,11} Research on such extraterrestrial construction materials also holds significant implications for developing new-generation high-performance building materials for extreme climates or special protective engineering on Earth.

To address the substantial cost of Earth-Moon transportation, in situ resource utilization (ISRU) has become a fundamental strategy for achieving sustainable extraterrestrial construction.^{12,13} Analyses of lunar regolith indicate that the lunar mare region is rich in basaltic minerals, with a chemical composition highly similar to that of terrestrial basalt.^{14–16} This has prompted researchers to explore the melting and drawing of such materials to produce basalt fiber (BF).^{17–19} As a green inorganic fiber, BF has been widely applied in various fields, including construction reinforcement and composite materials on Earth,^{20,21} exhibiting comprehensive advantages such as high strength, high modulus, and corrosion resistance. More importantly, recent studies have confirmed the feasibility of producing BF using simulated lunar regolith,²² and research has revealed that under simulated extreme lunar temperature cycling conditions, basalt fiber can effectively enhance the strength of geopolymers through a bridging mechanism, thereby playing a role in in situ lunar base construction.²³ However, this composite material system still struggles to address the challenges of electromagnetic shielding and active thermal management. Therefore, effective surface engineering is necessary to functionally upgrade BF, altering its inherently insulating properties and making it a highly promising structural-functional integrated substrate within an ISRU-inspired context.

In the fields of construction materials and composites, surface metallization is a mature strategy for imparting functionality to nonconductive substrates.²⁴ Among the techniques, electroless plating and electroplating have been extensively studied for their ability to form uniform, dense, and well-adhered metal coatings on complex surfaces.^{25,26} Our previous research successfully optimized a nickel–copper bilayer metallization process on high-performance organic PBO fibers, achieving excellent electrical conductivity and electromagnetic shielding effectiveness.²⁷ However, when applying this surface engineering technique to BF, a critical issue that cannot be overlooked is the mismatch in thermophysical properties between the materials: as an inorganic silicate fiber, BF has a coefficient of thermal expansion (CTE) of approximately $6 \times 10^{-6}/\text{K}$,²⁸ while that of metallic Cu is as high as $17.4 \times 10^{-6}/\text{K}$.²⁹ Under the extreme temperature alternation environment on the lunar surface, with a variation of up to 310 °C, this significant CTE mismatch induces substantial cyclic thermal stress at the fiber-metal interface. Without effective interfacial control strategies, such thermal stress can readily lead to catastrophic failures, including coating peeling and interfacial debonding, resulting in the complete loss of material functionality. Notably, the introduction of a Ni layer (CTE approximately $12.9 \times 10^{-6}/\text{K}$)²⁹ precisely creates a gradient transition in CTE between BF and Cu, theoretically capable of significantly mitigating interfacial thermal stress concentration and thus suppressing catastrophic interface failure.^{30,31} Although this process holds promise for both functionality and interfacial control, its application to ISRU-based substrates intended for extra-

terrestrial construction, combined with systematic evaluation of its long-term durability, microscopic damage evolution, and functional failure mechanisms under simulated space extreme thermal cycling, remains a research gap. Existing studies predominantly focus on initial performance characterization at room temperature or under single environmental factors,^{32,33} lacking systematic investigation under wide-range temperature cycling, a typical thermal load on the Moon.

In view of this, this study aims to address the critical gap between the aforementioned material properties and validation under extreme environments. The core effort is directed toward rigorous engineering validation: transplanting the previously optimized Ni–Cu bilayer metallization process onto a basalt fiber substrate, utilizing the gradient buffering effect of the Ni layer to resolve the interfacial thermal mismatch issue, and conducting accelerated tests simulating the extreme temperature environment of the lunar surface. By systematically characterizing the electromagnetic shielding effectiveness, joule heating performance, electrical resistance stability, and microstructure of the material before and after cold-thermal shock, the objective is to reveal the functional degradation patterns and physical failure mechanisms, specifically investigating whether, under thermal stress, the material's failure mode transitions from direct interfacial cracking and peeling to gradual microstructural degradation. To ensure the rigor and generalizability of the study's conclusions and to thoroughly analyze the influence of the substrate material, glass fiber is introduced as a parallel control. This design serves a dual purpose: first, to verify the stability, reliability, and universality of the previously optimized metallization process across different silicate fiber substrates, demonstrating its potential as a reliable surface functionalization method; second, by systematically comparing the performance of the two types of substrate fibers, which differ in microstructure, surface chemistry, and thermophysical properties, after metallization, it is possible to effectively decouple and quantify the contributions of substrate characteristics and metal coating to the final composite's multifunctionality. This allows for a more scientific and precise elucidation of the unique behavior mechanisms of the basalt fiber-metal interface under thermal stress. A full ISRU implementation would ultimately require fibers melt-drawn from lunar regolith or its simulants; the present study uses commercial terrestrial basalt fiber as a well-characterized analog to first establish the metallization process and evaluate its durability under extreme thermal cycling. On this basis, this research not only provides indispensable experimental evidence and design guidance for future multifunctional composites intended for in situ lunar construction but also offers direct reference value for developing multifunctional building materials suitable for extreme climates or specialized protection requirements on Earth.

2. EXPERIMENTAL SECTION

2.1. Materials

Basalt fiber plain weave fabric was purchased from Jiaxing Anlide Construction Technology Co., Ltd., and glass fiber (GF) plain weave fabric was purchased from Shandong Senhong Engineering Materials Co., Ltd. Both fabrics were used as received without further weaving or fiber separation. The pretreatments and metallization processes were applied directly to the fabric pieces. Silver nitrate (AgNO_3) was obtained from Shanghai Fine Chemical Materials Research Institute. Stannous chloride ($\text{SnCl}_2 \cdot \text{H}_2\text{O}$) and nickel sulfate ($\text{NiSO}_4 \cdot 6\text{H}_2\text{O}$)

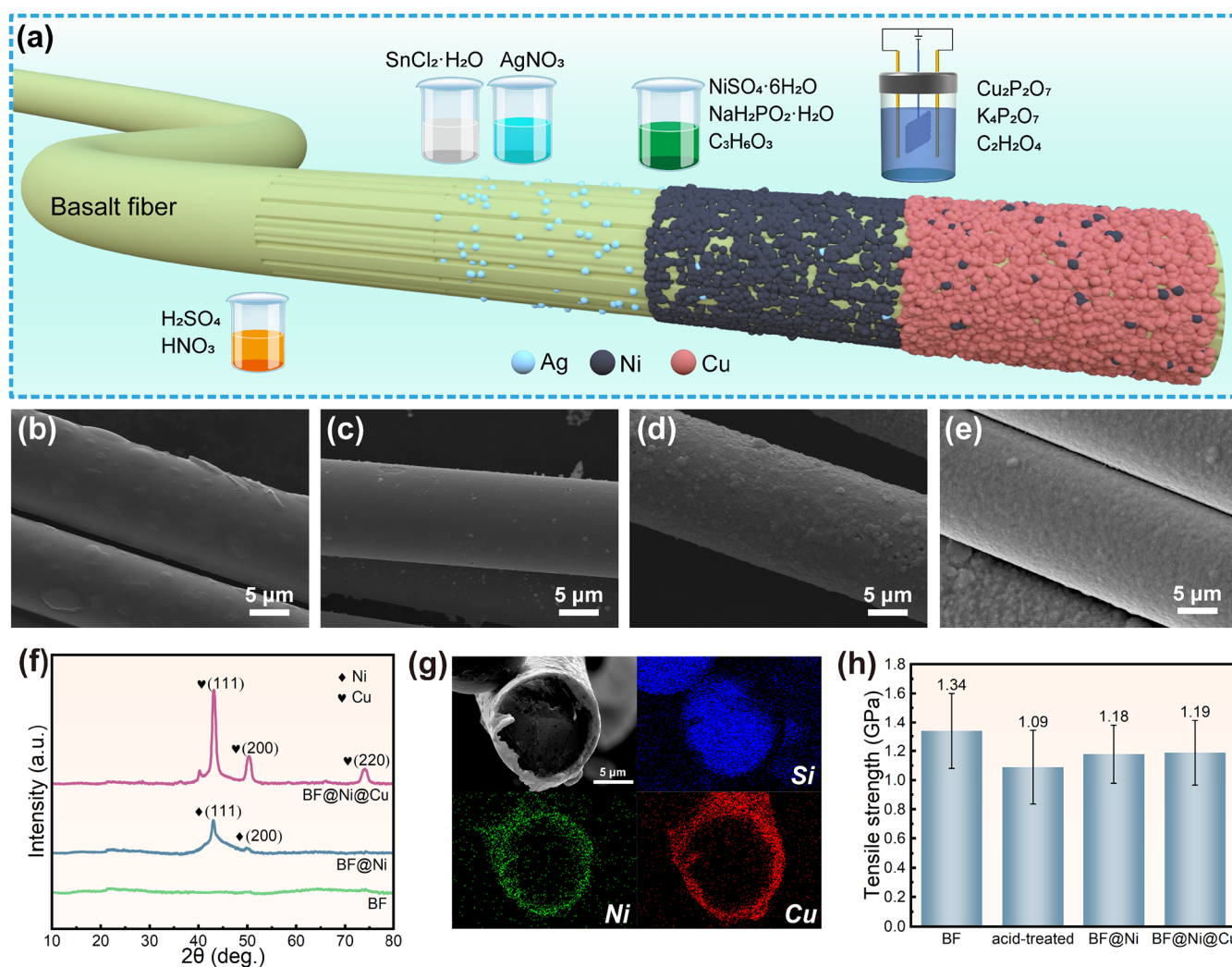


Figure 1. Preparation and structural characterization of surface metallized basalt fiber (BF). (a) Schematic diagram of the Ni/Cu plating process on the basalt fiber surface; (b–e) SEM images of BF after acid treatment, after activation, after nickel plating, and after copper plating; (f) XRD patterns of BF after Ni/Cu plating; (g) Cross-sectional SEM image and elemental mapping of BF@Ni@Cu; (h) Tensile strength of basalt fiber at different stages during the metallization process.

were purchased from Tianjin Aopusheng Chemical Co., Ltd. Sodium hypophosphite ($\text{NaH}_2\text{PO}_2 \cdot \text{H}_2\text{O}$) was purchased from Sinopharm Chemical Reagent Co., Ltd. Lactic acid ($\text{C}_3\text{H}_6\text{O}_3$) was purchased from Tianjin Fuyu Fine Chemical Co., Ltd. Copper pyrophosphate ($\text{Cu}_2\text{P}_2\text{O}_7$), anhydrous oxalic acid ($\text{C}_2\text{H}_2\text{O}_4$), and potassium pyrophosphate ($\text{K}_4\text{P}_2\text{O}_7$) were purchased from Shanghai Ean Chemical Technology Co., Ltd. PDMS and its curing agent were purchased from Suzhou Haitisi Electronic Technology Co., Ltd. All reagents were of analytical reagent grade and were not further purified.

2.2. Preparation of Basalt Fiber@Ni

First, the basalt fiber fabric was soaked in acetone for 30 min to remove the surface sizing layer and impurities. Subsequently, the desized basalt fiber fabric was placed in a mixed solution of 200 mL/L H_2SO_4 and 200 mL/L HNO_3 and treated with ultrasonic oscillation (power: 100 W, frequency: 40 kHz) for 30 min to obtain the modified BF fabric. Next, under a water bath condition of 40 °C, the modified fiber fabric was immersed in a sensitization solution containing 10 g/L $\text{SnCl}_2 \cdot \text{H}_2\text{O}$ for 30 min. Afterward, at the same water bath temperature, the sensitized BF was immersed in an activation solution for 30 min. The activation solution was a 5 g/L AgNO_3 solution, into which a 10% NH_4OH solution was added dropwise until the solution became transparent. Finally, the pretreated basalt fiber fabric was placed in an electroless nickel plating solution and

plated for 30 min in a water bath at 80 °C. The composition of the electroless nickel plating solution was as follows: $\text{NiSO}_4 \cdot 6\text{H}_2\text{O}$ 35 g/L, $\text{NaH}_2\text{PO}_2 \cdot \text{H}_2\text{O}$ 35 g/L, and lactic acid 30 mL/L.³⁴ After the plating process, the sample was rinsed three times with deionized water and then placed in a 60 °C baking oven for drying to obtain the nickel-plated basalt fiber fabric (BF@Ni).

2.3. Preparation of Basalt Fiber@Ni@Cu

BF@Ni served as the cathode, and a Cu rod served as the anode. The connected cathode and anode were immersed in the electroplating solution, and the power supply was turned on for electroplating. The electroplating solution consisted of $\text{Cu}_2\text{P}_2\text{O}_7$ 25 g/L, $\text{K}_4\text{P}_2\text{O}_7$ 200 g/L, and anhydrous $\text{C}_2\text{H}_2\text{O}_4$ 25 g/L, with the pH adjusted to 8. After electroplating, the sample was rinsed three times with deionized water and then placed in a 60 °C baking oven for drying to obtain BF@Ni@Cu.

2.4. Preparation of Basalt Fiber@Cu

BF@Cu was prepared through a sequential process of chemical plating followed by electroplating. The pretreatment step in the chemical plating process was the same as that for BF@Ni. The composition of the copper chemical plating bath was: $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ 20 g/L, 37% HCHO 13 mL/L, Na_2EDTA 15 g/L, $\text{KNaC}_4\text{H}_4\text{O}_6 \cdot 4\text{H}_2\text{O}$ 20 g/L, NaOH 12 g/L. After the chemical plating of copper was completed, electroplating of copper was performed using the same

process as that used in the preparation of BF@Ni@Cu, and the total coating thickness was maintained to match the total thickness of the Ni/Cu layer in BF@Ni@Cu. After electroplating, the sample was rinsed three times with deionized water and then placed in a 60 °C baking oven for drying to obtain BF@Cu.

2.5. Annealing and Polydimethylsiloxane Encapsulation

BF@Ni@Cu was placed in a muffle furnace and annealed in air at different temperatures for 1 h to obtain BF@Ni@Cu-X (where X denotes the temperature). Subsequently, polydimethylsiloxane (PDMS) prepolymer and curing agent were mixed in a beaker at a mass ratio of 10:1. After stirring to achieve uniform mixing, bubbles were removed by ultrasonic oscillation. The woven fabric was placed into a self-made mold, and the prepared PDMS mixture was injected. The mold was then transferred to an oven and cured at 90 °C for 2 h, ultimately yielding BF@Ni@Cu-X/PDMS. The metallization process for the glass fiber fabric substrate was consistent with that for the basalt fiber fabric.

2.6. Characterization

The samples were characterized by using a field-emission scanning electron microscope (Quanta-450-FEG) and EDS (X-MAX50). The absorption peak vibration was tested by Nicolet is50 Fourier transform infrared spectrometer (FT-IR, Thermo Fisher Scientific, China) in the range of 4000~400 cm^{-1} . X-ray diffraction (XRD) analysis was characterized from 10 to 80° by DX-2700BH X-ray diffractometer (Haoyuan Instrument, China) with a scanning speed of 6°·min⁻¹ and an operating voltage of 40 kV. The EMI shielding performance was measured by N5232A network analyzer (AGILENT, USA) at 8.2~12.4 GHz (X band). Single fiber tensile strength was measured by LLY-06EDC Single fiber strength tester (Laizhou, China). Measure the surface temperature of woven fabric using HM-TPE09se visual thermometer (Hikvision, China).

The electromagnetic interference shielding capability of the metallized fiber fabric in the 8.2~12.4 GHz (X-band) was characterized by the scattering parameters (S_{11} , S_{21}) recorded using a vector network analyzer. The power coefficients (R , A , T), as well as the reflection loss (SER), absorption loss (SEA), and total shielding effectiveness (SET), can be calculated from the scattering parameters using the following formulas:

$$R = |S_{11}|^2 = |S_{22}|^2 \quad (1)$$

$$T = |S_{12}|^2 = |S_{21}|^2 \quad (2)$$

$$A = 1 - R - T \quad (3)$$

$$\text{SER} = -10 \log(1 - R) \quad (4)$$

$$\text{SEA} = -10 \log(T/(1 - R)) \quad (5)$$

$$\text{SET} = \text{SER} + \text{SEA} \quad (6)$$

3. RESULTS AND DISCUSSION

3.1. Preparation and Properties of Basalt Fiber@Ni@Cu

As shown in Figure 1a, a schematic diagram of the preparation process for basalt fiber fabric metallization is presented. Figure 1b–e display the surface morphologies of basalt fiber at the corresponding steps during the metallization process. It can be observed from the images that after acid treatment, grooves appeared on the originally smooth surface of the basalt fiber (Figure S1), as shown in Figure 1b. These grooves microscopically facilitate the enhancement of physical interlocking between the coating and the basalt fiber to a certain extent. Furthermore, the acid treatment also enhances the surface activity of the fiber,^{35,36} which is beneficial for subsequent reactions. Following the activation step, Ag^+ was reduced to Ag by Sn^{2+} and deposited on the basalt fiber surface (Figure 1c). Ag acts as a catalytic active site, causing Ni nanoparticles to

preferentially deposit around them during electroless nickel plating,³⁷ until Ni completely coats the fiber surface (Figure 1d), yielding BF@Ni. Finally, a Cu layer was electrodeposited onto the Ni layer (Figure 1e), resulting in BF@Ni@Cu. A macroscopic photograph of a representative 7 cm × 7 cm BF@Ni@Cu fabric is shown in Figure S2a. The fabric exhibits a uniform copper-colored metallic appearance across the entire area, with no visible macroscopic defects or uncoated spots. To evaluate flexibility, the fabric was manually bent and folded; as shown in Figure S2b, the metallic coating remains intact without cracking or peeling, indicating good mechanical robustness for flexible applications. The metallization process for glass fiber fabric was consistent with that for basalt fiber fabric, and the SEM images of each step are shown in Figure S3. Compared with basalt fiber, the reason for the presence of pores on the coating surface of metallized glass fiber is the high surface chemical inertness, uneven distribution of catalytically active centers, poor wettability, and weak interfacial bonding of the GF surface. These factors lead to low nucleation density and discontinuous deposition during electroless nickel plating, accompanied by hydrogen gas retention and internal stress. Precisely due to the discontinuity of the underlying nickel layer, the pores cannot be remedied by electroplating.

Figure 1f presents the XRD patterns of BF, BF@Ni, and BF@Ni@Cu. BF exhibited no obvious characteristic peaks, showing an amorphous characteristic.^{14,38} In the pattern of the nickel-plated basalt fiber, characteristic peaks of Ni (111) and (200) appeared at 45 and 51.8°. In the pattern of BF@Ni@Cu, in addition to the Ni characteristic peaks, Cu characteristic peaks were present at 43.3° (111), 50.4° (200), and 74.1° (220), indicating the successful deposition of the nickel–copper bimetallic layer. Figure 1g shows the cross-sectional SEM image and elemental mapping of the nickel–copper plated basalt fiber. From the image, it can be observed that the bimetallic layer on the fiber surface is completely and densely coated, with a thickness of approximately 1 μm . The coating is tightly bonded to the fiber substrate, and the overall cross-section exhibits a clear multilayer structure. Notably, the Ni layer, which has a moderate coefficient of thermal expansion, is precisely positioned between the basalt fiber and the Cu layer, forming a gradient transition structure. This design lays the structural foundation for subsequent interfacial stress regulation under extreme temperature environments.³⁹

The tensile strength of basalt fiber at different stages is shown in Figure 1h. The tensile strength of pristine basalt fiber, after roughening, after nickel plating, and after copper plating was tested, respectively. The results showed that after acid treatment, the fiber strength decreased from 1.34 to 1.09 GPa. This phenomenon occurs because acid treatment dissolves metal oxides on the fiber surface, disrupts the stability of the silicon–oxygen network, leading to the formation of micropores and cracks on the surface that act as stress concentration points. Defects may also penetrate along the fiber axis, weakening weak bond regions within the amorphous network, causing irreversible bulk damage. After Ni/Cu deposition, the metal layer covered the surface defects generated by the acid treatment, reducing stress concentration. Moreover, the mechanical interlocking and chemical bonding between the coating and the fiber can enhance the interfacial shear strength, facilitating stress transfer from the fiber to the coating.⁴⁰ Consequently, the tensile strength increased to 1.19 GPa. While acid roughening enhances coating adhesion, it compromises the intrinsic strength of the fiber; metal coating

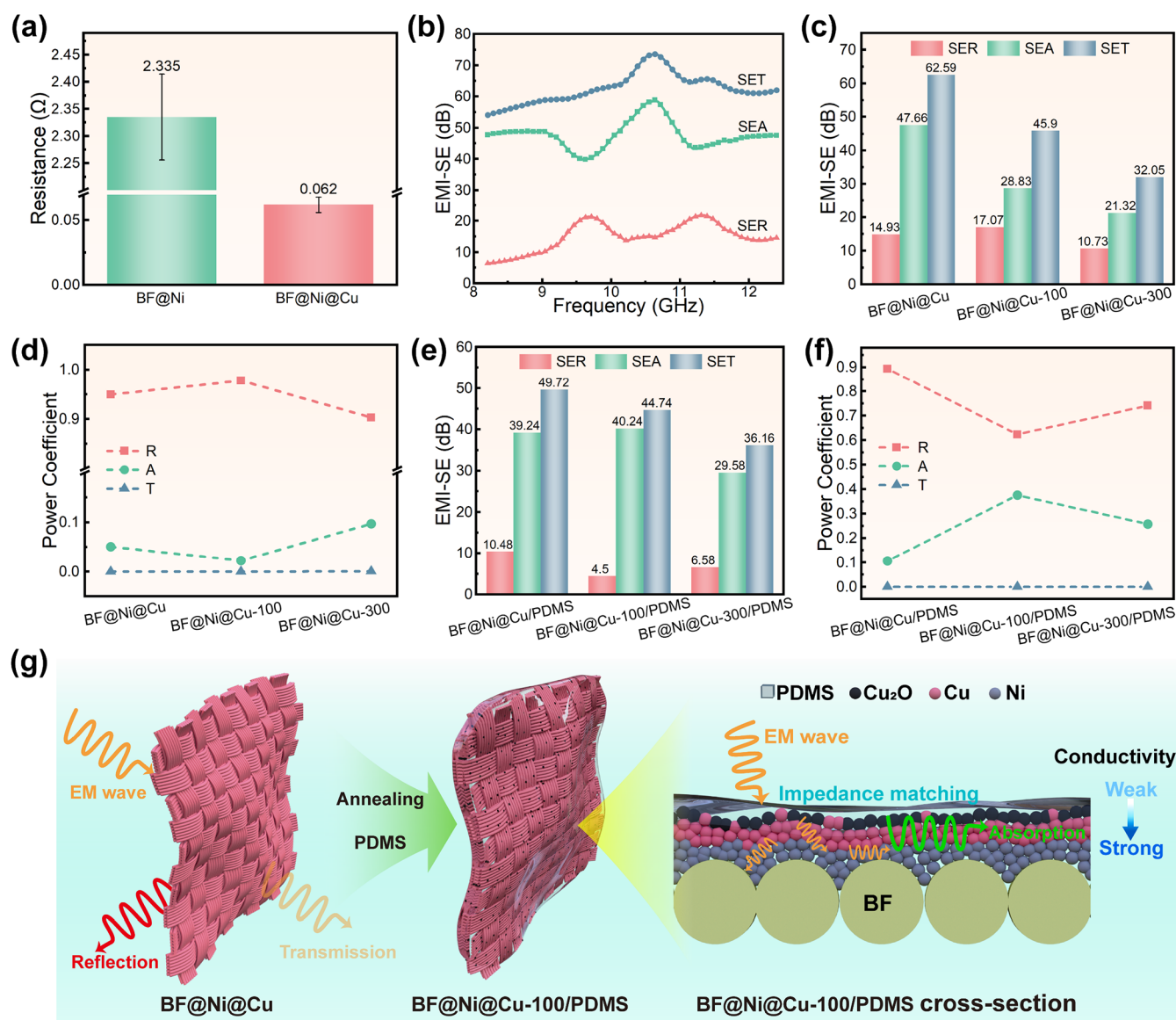


Figure 2. Electromagnetic interference shielding performance and shielding mechanism modulation of BF@Ni@Cu. (a) Surface resistance of BF@Ni and BF@Ni@Cu; (b) Electromagnetic interference shielding effectiveness curves of BF@Ni@Cu; (c) SEA and SER values of BF@Ni@Cu, BF@Ni@Cu-100, and BF@Ni@Cu-300; (d) R, A, and T values of BF@Ni@Cu, BF@Ni@Cu-100, and BF@Ni@Cu-300; (e) SEA and SER values of BF@Ni@Cu/PDMS, BF@Ni@Cu-100/PDMS, and BF@Ni@Cu-300/PDMS; (f) R, A, and T values of BF@Ni@Cu/PDMS, BF@Ni@Cu-100/PDMS, and BF@Ni@Cu-300/PDMS; (g) Electromagnetic wave shielding mechanisms of BF@Ni@Cu fabric and BF@Ni@Cu-100/PDMS fabric composite.

can repair surface defects but cannot reverse internal damage. Therefore, moderating the pretreatment process represents a direction for future improvement.

3.2. Electromagnetic Interference Shielding Performance of Basalt Fiber@Ni@Cu

After the deposition of the nickel–copper bimetallic layer on the surface of the BF fabric, the electrical conductivity was significantly enhanced. The resistance measured between any two points 3 cm apart on the fabric decreased from being originally infinite to the order of ohms. Specifically, the resistance of the basalt fiber fabric after nickel plating was 2.335 Ω , and after the introduction of Cu via electroplating, the resistance further decreased to 0.062 Ω . Benefiting from this excellent electrical conductivity, the metallized fabric also exhibited outstanding electromagnetic shielding performance. In the X-band, the electromagnetic interference shielding

effectiveness of both BF@Ni and GF@Ni reached 49 dB (Figure S4). After copper plating, the electromagnetic interference shielding performance of BF@Ni@Cu was further enhanced to 62.59 dB (Figure 2b), which corresponds to an attenuation of over 99.9999% of the incident electromagnetic wave energy. When an electromagnetic wave impinges on the fabric surface from the air medium, the high-concentration free electrons within the highly conductive metal layer rapidly form an induced current driven by the alternating electromagnetic field, subsequently generating a secondary electromagnetic field with an opposite phase to the incident wave.⁴¹ The macroscopic manifestation of this microscopic process is that the metallized surface exhibits an extremely low wave impedance, creating a strong impedance mismatch with the wave impedance of air. Consequently, the majority of the incident wave energy is reflected back into the original space at

the interface. Meanwhile, the small portion of electromagnetic waves that penetrate the material induces eddy currents within the conductive network, which are converted into joule heat loss,⁴² further contributing to absorption attenuation, thereby achieving electromagnetic wave shielding.

As mentioned above, the shielded electromagnetic waves do not disappear but are reflected back into free space, implying that these waves still cause secondary pollution. To better meet the requirements for green electromagnetic shielding, improvements were made to BF@Ni@Cu to enhance the absorption-dominated shielding mechanism. BF@Ni@Cu was placed in a muffle furnace and annealed at 100 and 300 °C for 1 h, respectively. The electromagnetic interference shielding performance of the annealed nickel–copper plated basalt fiber fabric is shown in Figure 2c. As the annealing temperature increased, the SE value of BF@Ni@Cu gradually decreased to 45.9 and 32.05 dB, respectively. Concurrently with the sacrifice of SE value, the absorption coefficient A of the fabric fluctuated slightly, as shown in Figure 2d. This phenomenon originates from the annealing-induced microstructural evolution:⁴³ annealing at 100 °C primarily causes slight surface oxidation and grain relaxation, forming a thin oxide layer on the surface. This layer slightly reduces the overall electrical conductivity, leading to a decreased SE value, but the additional interfacial impedance mismatch introduced causes the reflection coefficient R to increase slightly to 0.977. High-temperature annealing at 300 °C leads to severe surface oxidation and interfacial diffusion, forming a thick and dense insulating oxide layer and potentially generating microcracks. This disrupts the continuity of the conductive network, resulting in a decrease in the intrinsic conductivity of the material, causing a significant loss in electromagnetic wave shielding capability. On the other hand, the extremely low electrical conductivity increases the absolute value of its wave impedance, relatively weakening the degree of impedance mismatch with air, causing the reflection coefficient R to drop to 0.903.

To further modulate the shielding mechanism, a layer of PDMS was encapsulated onto the annealed fabric. For the highly conductive BF@Ni@Cu and BF@Ni@Cu-100, the PDMS layer acts as an impedance transition layer between the air and the metal surface. Its dielectric constant lies between that of air and metal, alleviating the sharp impedance mismatch and allowing more electromagnetic waves to enter the composite system rather than being directly reflected.⁴⁴ The incoming electromagnetic waves undergo multiple reflections within the PDMS layer and are ultimately absorbed by the metal layer. However, this process does not fully compensate for the total attenuation loss caused by the reduction in strong reflection, hence the SE value decreases (Figure 2e). For BF@Ni@Cu-300, which inherently possesses very poor shielding capability, the impedance matching effect of PDMS also reduces the R value (Figure 2f). On the other hand, the PDMS layer itself and the newly formed interfaces provide additional multiple reflection and dielectric loss pathways for electromagnetic waves. These additional loss mechanisms contribute extra attenuation, leading to a slight increase in the overall SE value.

The mechanism behind the outstanding electromagnetic interference shielding performance exhibited by the metallized fabric is illustrated in Figure 2g. The shielding mechanism of the BF@Ni@Cu fabric is primarily reflection-dominated, as described above. Notably, the BF@Ni@Cu-100/PDMS composite demonstrated the optimal overall performance:

while maintaining a good shielding effectiveness of 44.74 dB, its absorption coefficient increased to 0.38, indicating a significant enhancement in the absorption contribution. The underlying physical mechanism can be interpreted as an absorption-enhanced shielding structure rather than a purely absorption-dominated one. The advantages of this structure arise from synergy across multiple levels:⁴⁵ after mild annealing, BF@Ni@Cu-100 develops a thin oxide layer on its surface, which moderately reduces its surface conductivity, causing its intrinsic wave impedance to lie between that of an ideal conductor and an insulator, thus creating conditions for subsequent impedance matching. The subsequently encapsulated PDMS layer effectively bridges the high impedance of air and the modulated impedance of the substrate, enabling efficient injection of electromagnetic waves from air into the fabric rather than strong reflection at the interface.

The electromagnetic wave energy that successfully enters the composite then undergoes multiple loss processes, which can be categorized as follows: (i) Multiple reflections and scattering: Within the PDMS layer and at the PDMS/oxide/metal interfaces, electromagnetic waves experience repeated reflections. This prolongs their transmission path and increases the opportunity for energy dissipation per unit volume. Similar multiple-reflection mechanisms have been reported in PDMS-encapsulated metal-coated nonwoven fabrics, where the porous structure and the dielectric-metal interfaces collectively enhance absorption.^{46,47} (ii) Interfacial polarization loss: The presence of heterogeneous interfaces (PDMS/oxide, oxide/metal, and especially the Cu/Cu₂O interface formed during mild annealing) gives rise to interfacial polarization under an alternating electromagnetic field. The periodic build-up and relaxation of space charges at these interfaces convert electromagnetic energy into thermal energy, contributing significantly to the dielectric loss. In particular, the Cu/Cu₂O heterojunction has been identified as a key source of enhanced interfacial polarization in copper-based composite shields.⁴⁶ (iii) Conduction loss: The residual metallic conductive network, including the underlying Ni–Cu layer, allows induced eddy currents to flow, converting electromagnetic energy into ohmic heat. This mechanism dominates in the highly conductive pristine BF@Ni@Cu but also contributes, to a lesser extent, in the annealed and PDMS-encapsulated sample.

Another benefit of using PDMS encapsulation is that it provides an excellent physical barrier and a chemically inert protective layer for the easily oxidized Cu layer on the BF@Ni@Cu surface. Because PDMS possesses excellent hydrophobicity, chemical stability, and low gas permeability, it effectively prevents water vapor, oxygen, and corrosive media from coming into contact with the internal metal coating, thereby significantly improving the environmental stability of the composite material. Figure S5 shows the change in electromagnetic interference (EMI) shielding performance of the PDMS-encapsulated BF@Ni@Cu sample after 72 h in an accelerated aging environment (85 °C with high humidity), which decreased from 49.72 to 44.27 dB—a decline of less than 6 dB. This result strongly demonstrates the effectiveness of the PDMS encapsulation layer in ensuring the long-term oxidation resistance of the composite fabric and maintaining the stability of its EMI shielding performance.

GF@Ni@Cu exhibited similarly excellent electromagnetic interference shielding performance (Figure S6), and the trends in SE value and reflection coefficient R after annealing and PDMS encapsulation were consistent with those of BF@Ni@

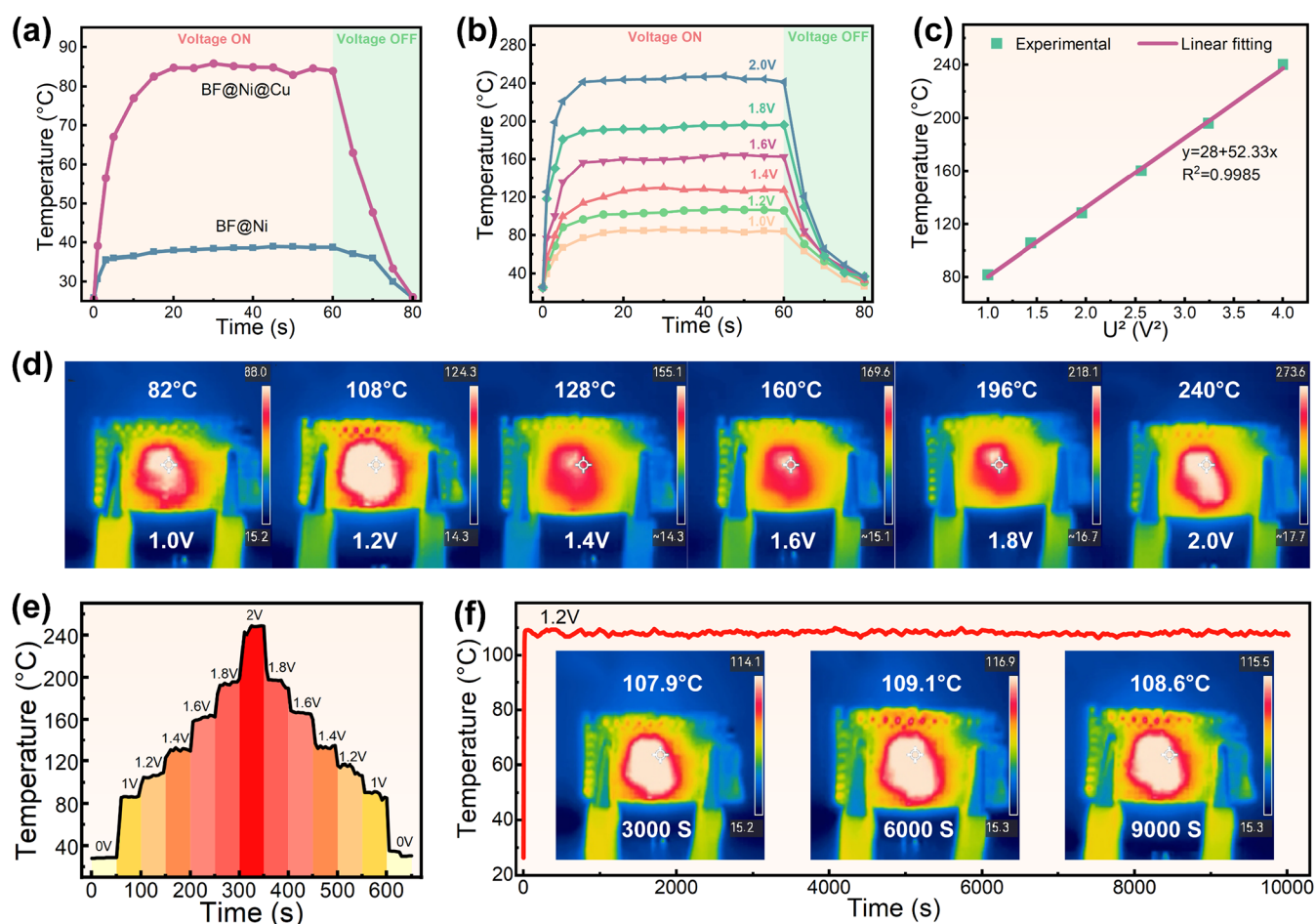


Figure 3. Electrothermal conversion performance of BF@Ni@Cu. (a) Surface temperature of BF@Ni and BF@Ni@Cu at 1.0 V; (b) Surface temperature of BF@Ni@Cu under different voltages; (c) Linear fitting of temperature variation versus the square of the applied voltage; (d) Thermal images of BF@Ni@Cu under different voltages; (e) Surface temperature of BF@Ni@Cu under stepwise voltage; (f) Long-term surface temperature stability of BF@Ni@Cu at 1.2 V.

Cu. Again, GF@Ni@Cu-100/PDMS demonstrated the optimal overall performance, with a shielding effectiveness of 33.35 dB and an absorption coefficient A of 0.38. This indicates that the success of the shielding modulation mechanism is not coincidental. It represents a strategy combining intrinsic surface modulation with external impedance design, achieving an active transition from the traditional reflection-dominated shielding mode toward a more desirable low-reflection, high-absorption mode. This structure reduces secondary electromagnetic pollution while maintaining excellent overall shielding capability, holding significant application potential in electromagnetic compatibility and stealth technologies.

3.3. Joule Heating Performance of Basalt Fiber@Ni@Cu

The high electrical conductivity of BF@Ni@Cu makes it an excellent joule heater, holding broad application prospects in construction and multifunctional composite material fields.⁴⁸ Figure 3a illustrates the surface temperature of BF@Ni and BF@Ni@Cu, with dimensions of 2 cm × 3 cm, under a voltage of 1 V. The fabric samples were cut into dimensions of 2 cm × 3 cm. Copper alligator clips were used as electrodes and clamped onto both ends of each sample; each clip gripped approximately 5 mm of the fabric edge, and the distance between the two clips was 2 cm. Both types of fabric rapidly heated up within 10 s after voltage application and then

remained stable. The stable surface temperature of BF@Ni was 36 °C. Benefiting from the high electrical conductivity and thermal conductivity of Cu, BF@Ni@Cu exhibited a superior stable surface temperature of 82 °C, indicating that the introduction of Cu has a significant positive impact on the electrothermal conversion performance of the fabric. Figure 3b shows the temperature variation of BF@Ni@Cu under different voltages. As the applied voltage increased, the surface temperature of the sample correspondingly increased. When the voltage was increased to just 2 V, the surface temperature of the composite fabric reached as high as 240 °C and remained stable. The high steady-state temperature achieved at 2 V does not imply continuous operation at such extremes; rather, it provides a wide voltage-tunable heating range, enabling both routine low-temperature thermal management at reduced voltage and emergency high-power heating within a single material. Owing to its inherent heat resistance, BF as a substrate can fully withstand the high heat generated by the surface metal layer. Within the voltage range of 1 to 2 V, the temperature of the fabric increased to its respective stable value within 10 s and returned to room temperature within 20 s after the power was disconnected.

The temperature of BF@Ni@Cu under voltages of 1–2 V exhibited a linear relationship with the U^2 value (Figure 3c), which is a direct manifestation of Joule's law under conditions

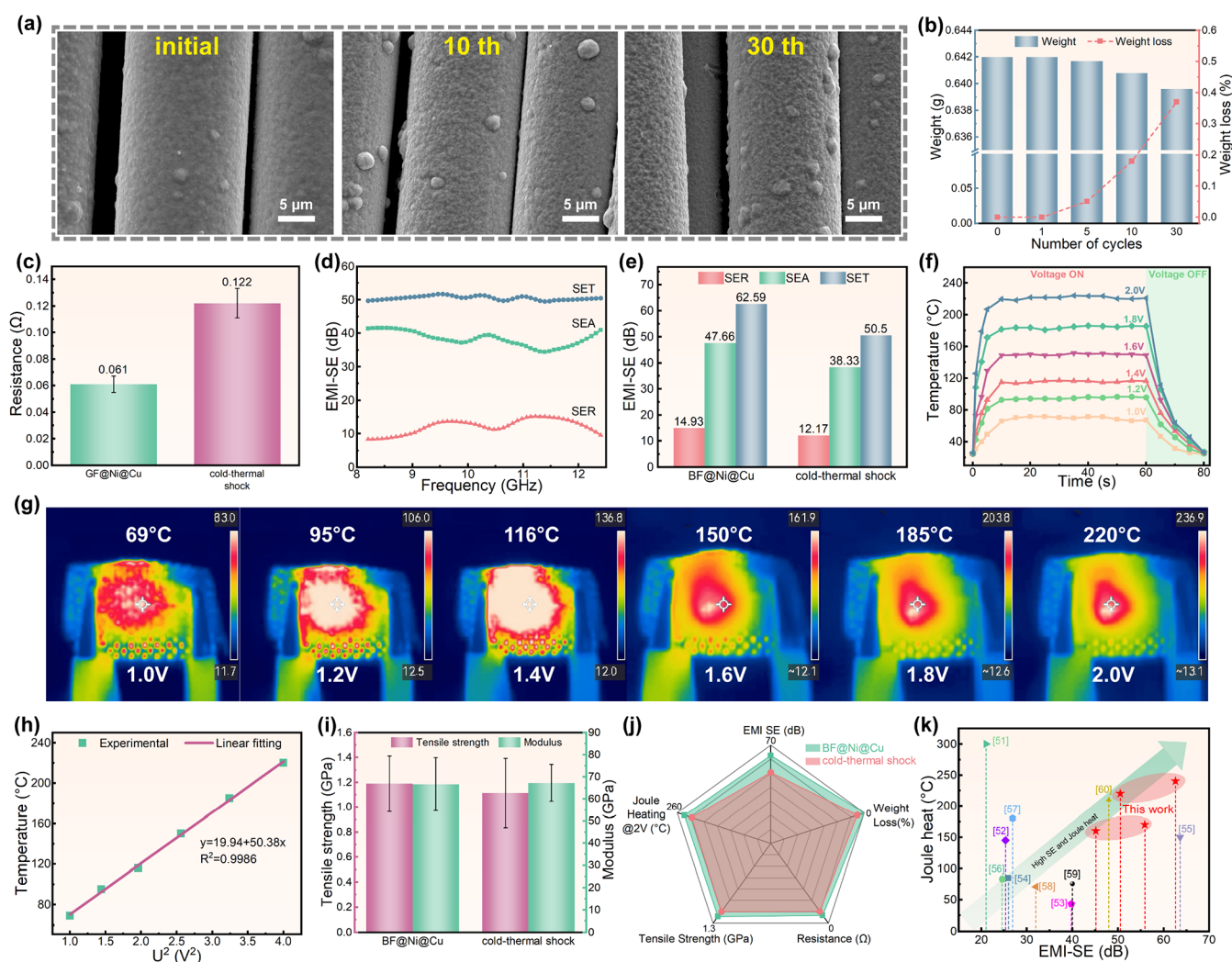


Figure 4. Performance changes of BF@Ni@Cu after cold-thermal shock cycling. (a) SEM images; (b) Weight loss rate; (c) Resistance change; (d) Electromagnetic interference shielding effectiveness curves; (e) SE value change; (f) Electrothermal conversion performance; (g) Thermal images of BF@Ni@Cu under different voltages; (h) Linear fitting of the square of the applied voltage versus temperature change; (i) Tensile strength and modulus; (j) Comparison of key properties of BF@Ni@Cu before and after cold-thermal shock cycling; (k) Performance comparison between this study and other studies.

of relatively stable material resistance and thermal equilibrium. This demonstrates the purity and controllability of the heating mechanism, indicating that it is a stable and reliable electric heater. The surface temperature of the fabric under different voltages is shown in Figure 3d. When a stepwise increasing voltage was applied to BF@Ni@Cu, the fabric temperature responded rapidly, exhibiting high electrothermal sensitivity (Figure 3e), enabling precise temperature control. Furthermore, the stability of joule heating output during long-term use is crucial. BF@Ni@Cu demonstrated excellent stability and reliability for long-term application. Under a voltage of 1.2 V, the fabric temperature stabilized near 88 °C and exhibited only minimal fluctuations over 10,000 s, showing virtually no change (Figure 3f). This long-term stability is attributed to the good coating adhesion, the moderate operating temperature, and the excellent thermal stability of the basalt fiber substrate.

The metallized glass fiber also exhibited excellent electrothermal conversion performance (Figure S7). However, due to the lower thermal conductivity of the glass fiber substrate compared to basalt fiber, the internal heat conduction and surface heat dissipation efficiency were slightly lower, resulting

in a lower surface temperature than that of BF@Ni@Cu. Nevertheless, the maximum surface temperature of GF@Ni@Cu under a voltage of 2 V could reach 170 °C, and it also conformed to Joule's law, still demonstrating its potential as a thermal management material.

3.4. Performance of Basalt Fiber@Ni@Cu under Simulated Lunar Surface Temperature

To evaluate the performance reliability of BF@Ni@Cu under lunar conditions, cold-thermal shock cycle tests simulating the extreme temperature environment of the Moon were conducted. Specifically, BF@Ni@Cu was placed in an oven at 130 °C for 10 min, then quickly removed and immersed in liquid nitrogen for 10 min; this constituted one cold-thermal shock cycle, and a total of 30 such cycles were performed. The selected temperatures (130 and −196 °C) fully encompass the temperature fluctuation range on the lunar surface, and the test results provide valuable reference for assessing whether the fabric can operate stably on the lunar environment temperature.

Figure 4a shows SEM images of BF@Ni@Cu after the 10th and 30th cold-thermal shock cycles. It can be observed that as

the number of cold-thermal shock cycles increased, the surface roughness of the metallic coating on BF@Ni@Cu gradually increased, but no obvious cracks or coating peeling were observed. The core reason for this phenomenon lies in the significant difference in the coefficient of thermal expansion between the surface metal layer, especially the Cu layer, and the fiber substrate, leading to periodic thermal stress at the interface during repeated rapid heating and cooling.⁴⁹ Typically, such thermal stress can easily cause catastrophic failures such as interfacial debonding or coating cracking.⁵⁰ However, the Ni interlayer introduced in this material system forms a gradient transition in the coefficient of thermal expansion between BF and Cu, effectively alleviating the thermal stress concentration at the interface. Simultaneously, the strong chemical bonding formed between the coating and the fiber via electroless plating enables the interfacial bonding strength to withstand the damage from thermal stress. Consequently, the accumulated thermal stress did not trigger catastrophic interface failure but was gradually released by the coating in the form of plastic deformation, lattice distortion, and surface roughening. The progressive roughening is clearly visible, and the absence of cracks or peeling is the key evidence for the success of the thermal expansion gradient buffer strategy.

To verify the two mechanisms described above, a control sample (BF@Cu) without the Ni interlayer was subjected to cold-thermal shock cycles under the same conditions and for the same number of cycles to confirm the buffering effect of the Ni interlayer. Additionally, BF@Ni@Cu was treated with ultrasonication for 1 h (100 W, 40 kHz). This condition imposes strong mechanical stress on the coating-fiber interface. By observing the surface morphology of the coating after ultrasonic examination, it is possible to determine whether strong interfacial adhesion has formed between the fiber matrix and the coating. The test results are as follows: After 30 cycles, BF@Cu exhibited severe Cu layer cracking and delamination (Figure S8), confirming that the Ni interlayer is crucial for preventing catastrophic failure of the plating layer; after 1 h of ultrasonic treatment, the surface morphology of BF@Ni@Cu remained virtually unchanged, and testing of its surface resistance revealed only a slight increase of 0.051 Ω (Figure S9), which indirectly demonstrates that there is strong interfacial adhesion between the BF treated by chemical plating and electroplating and the metal plating layer as well as the Ni/Cu metal layer. The surface morphology of GF@Ni@Cu also exhibited phenomena similar to those observed for BF@Ni@Cu, with no cracking or peeling observed (Figure S10a), indicating that this material system successfully transformed the failure mode induced by thermal stress from traditional interfacial debonding or coating peeling to progressive microscopic damage. This is the key mechanism for maintaining structural reliability under extreme temperature alternation conditions.

The weight loss rate of the fabric after 30 cold-thermal shock cycles was only 0.4% (Figure 4b), further confirming that the overall structural integrity of the coating was well-maintained, withstanding the drastic temperature changes without significant shedding of coating fragments. The slight weight loss might be attributed to trace amounts of metal oxidation or minor brittle wear on the fiber surface at the low temperature of liquid nitrogen. The change in resistance of the fabric after cold-thermal shock cycling was also not significant, increasing only from 0.061 to 0.122 Ω (Figure 4c), approximately a one-

fold increase. The increase in resistance is primarily attributed to two factors: first, slight oxidation of the surface metal, and second, the increased coating roughness exacerbating surface scattering effects during electron conduction. Notably, although the resistance increased, no order-of-magnitude jump or open-circuit failure occurred, indicating that the continuity of the conductive network was maintained, representing an acceptable performance degradation. In contrast, the BF@Cu control sample showed a much more pronounced degradation, with its resistance rising from 0.079 to 0.498 Ω after 30 cycles (Figure S11a). Electromagnetic shielding effectiveness primarily depends on the electrical conductivity and macroscopic structural integrity of the material. Consequently, the increase in resistance directly caused the shielding performance of BF@Ni@Cu to drop from 62 to 50.5 dB (Figure 4d-e). For the BF@Cu control, the EMI shielding effectiveness dropped much more severely, from 51.15 to 28.99 dB (Figure S11b), further demonstrating that the Ni interlayer is crucial for maintaining functional performance under extreme cold-thermal shock.

After 30 cold-thermal shock cycles, the steady-state heating temperature of BF@Ni@Cu also decreased (Figure 4f-g). Specifically, after 30 cold-thermal shock cycles, the surface temperature of the composite fabric at a voltage of 1–2 V decreased by 10–20 $^{\circ}\text{C}$ compared to before the cold-thermal shock. According to Joule's law, under constant voltage conditions, the heating power is inversely proportional to the resistance. The increase in resistance resulted in less joule heat generated per unit time, thus lowering the steady-state temperature. Additionally, the increased surface roughness might enhance thermal radiation and convection, exacerbating heat dissipation, collectively contributing to the slight degradation in heating performance. Nevertheless, the electro-thermal conversion performance of the material still conformed to Joule's law, with a good linear fit between the square of the voltage and temperature ($R^2 = 0.9986$), as shown in Figure 4h. The changes in tensile strength and modulus of BF@Ni@Cu after cold-thermal shock are illustrated in Figure 4i. The tensile strength decreased slightly from 1.19 to 1.11 GPa, while the modulus remained essentially unchanged. The slight decrease in strength may stem from two aspects: first, the roughened coating enhances the surface stress concentration effect, which could act as initiation sites for microcracks; second, the cyclic thermal stress may cause microplastic damage in the interfacial region, slightly affecting load transfer efficiency. The absence of significant change in modulus indicates that the overall elastic behavior of the fiber substrate and coating was not substantially affected.

A comparison of key properties of the BF@Ni@Cu fabric before and after cold-thermal shock is shown in Figure 4j (Table S1). Based on the comprehensive analysis above, the material maintained high structural integrity and functional stability after experiencing 30 extreme cold-thermal shock cycles simulating the lunar temperature range. The core mechanism can be summarized as follows: through the gradient buffering effect of the Ni interlayer and the strong interfacial bonding design, the failure mode induced by thermal stress was successfully transformed from interfacial debonding or coating peeling to microscopic damage such as surface roughening and lattice distortion. The degradation in electrical, electromagnetic shielding, and joule heating performance primarily originated from the microscopic deterioration of electron transport pathways rather than

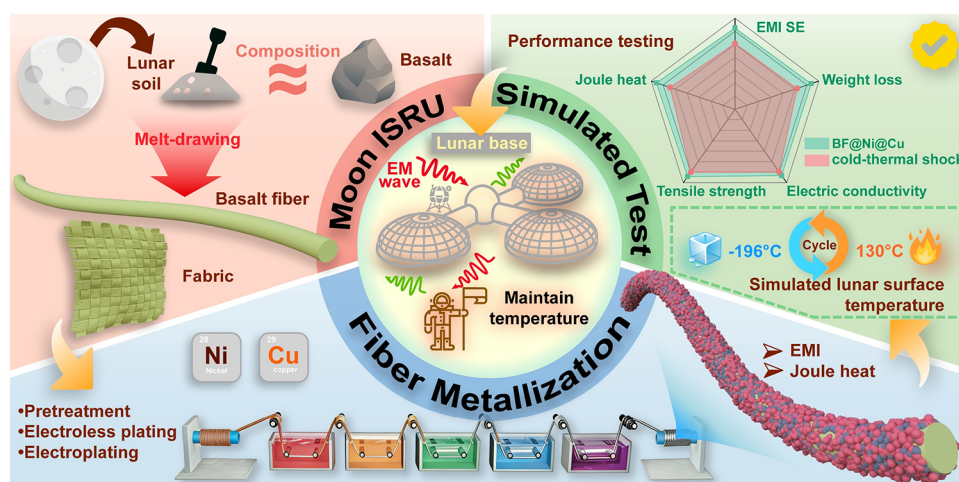


Figure 5. In-situ resource utilization-based multifunctional fiber for lunar base applications: from fabrication, performance validation, to comprehensive application.

macroscopic structural failure. The performance of GF@Ni@Cu also strongly corroborated this (Figure S10b–f). This composite material exhibits good resistance to temperature shock and environmental adaptability, making it suitable for functional applications such as lightweight electromagnetic shielding, thermal management, and sensing under the extreme diurnal temperature difference conditions on the lunar surface. Figure 4k presents a comparison of the EMI performance and joule heating performance of this study with other reported materials.^{51–60} The specific comparison parameters are provided in Table S2. Even after cold-thermal shock cycling, the performance of BF@Ni@Cu (SE value of 50.5 dB, steady-state surface temperature of 220 °C at 2 V) still holds advantages compared to other materials.

3.5. Application of Basalt Fiber@Ni@Cu under the Lunar Environment

Figure 5 illustrates the conceptual multifunctional application prospects of BF@Ni@Cu in future lunar base construction, based on an ISRU-inspired vision. While the current study uses commercial terrestrial basalt fiber as a proof-of-concept analog, and does not involve actual ISRU-derived fibers, it is envisioned that with future validation using lunar regolith simulants or actual regolith, continuous basalt fibers could be melt-drawn from lunar regolith through in situ resource utilization technology and woven into fabrics, providing a foundational substrate for subsequent functionalization. By employing a well-established process combining pretreatment, electroless plating, and electroplating, a nickel–copper composite metal layer is constructed on the fiber surface, successfully endowing it with electromagnetic interference shielding and electrothermal conversion capabilities, achieving the transformation from an insulating substrate to a conductive functional material. Furthermore, after cold-thermal shock testing simulating the extreme diurnal temperature range on the Moon, this metallized fiber maintains a stable structure and performance. Based on the performance validation described above, it can serve as a building shielding material or intelligent protective garment in future lunar base construction, providing electromagnetic protection and active thermal management for personnel and precision instruments, forming a conceptual technological loop from lunar regolith raw material to final

application, pending demonstration on actual ISRU-derived fibers.

It should be noted that the current cold-thermal shock tests do not account for other critical lunar environmental factors such as high vacuum, solar radiation, UV exposure, charged particles, or lunar dust abrasion. Future work will evaluate material performance under more realistic, combined environmental conditions. Additionally, to achieve all-weather, adaptive thermal management for lunar bases, future research could focus on expanding the functionality of this material system for the high-temperature lunar surface environment. For example, this could involve constructing a selectively high-reflectivity coating on the metallized surface to resist solar radiation, or compositing it with highly thermally conductive materials to enhance its heat dissipation efficiency into the lunar regolith. This would evolve the material from its current single function of active heating into a comprehensive solution with full thermal management capabilities.

4. CONCLUSIONS

Using commercial BF as a study for lunar basaltic materials, this study successfully constructed a robust nickel–copper composite metal layer on the surface of BF fabric through electroless nickel plating and electroplating processes, preparing a BF@Ni@Cu composite material that exhibits high electrical conductivity, excellent electromagnetic shielding effectiveness (62.59 dB), and significant joule heating performance (2 V, 240 °C). Annealing and PDMS encapsulation allow further modulation of its surface characteristics, enabling optimization of shielding performance and reflection behavior. To address the core challenge of fiber-metal interface thermal expansion mismatch under the extreme temperature alternation environment on the Moon, this study utilized the gradient buffering structure formed by the Ni interlayer between the BF substrate and the outer Cu layer, effectively alleviating interfacial thermal stress concentration. Cold-thermal shock cycle tests simulating the extreme temperature environment of the Moon demonstrated that the material maintained structural integrity after 30 cycles, without coating peeling or cracking, and exhibiting only surface roughening evolution. This phenomenon confirms that the gradient structure effectively improves the thermal stress-induced coating failure mode. Although electromagnetic shielding and

heating performance decreased slightly, they remained at relatively high levels. The slight performance degradation primarily originated from microscopic deterioration of electron transport pathways, demonstrating the reliable durability of this material system under extreme temperature shock. Repeating the same process on glass fiber yielded similar performance results, which provides that this metallization strategy possesses good process stability and substrate adaptability. And BF@Ni@Cu is a reliable material that maintains structural-functional integration under extreme temperature alternations. Admittedly, further comprehensive trade-off studies concerning mass, energy, and cost are still required to determine whether in-situ metallization of basalt fibers on the Moon is more practical than transporting lightweight prefabricated fabrics from Earth. Nonetheless, when combined with the potential for in situ fiberization of lunar regolith, BF@Ni@Cu still offers a promising material solution for the lightweight, multifunctional electromagnetic shielding and thermal management components required in future lunar bases, thereby demonstrating considerable engineering application value.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.6c07813>.

SEM images of pristine basalt fiber and glass fiber at different metallization stages; macroscopic images of BF@Ni@Cu; EMI shielding effectiveness curves of BF@Ni and GF@Ni in the X-band; optimization of EMI shielding performance and absorption coefficient of GF@Ni@Cu; electrothermal conversion performance of GF@Ni@Cu; SEM images and resistance of BF@Ni@Cu before and after ultrasonic treatment; morphology and performance changes of GF@Ni@Cu after cold-thermal shock cycling; morphology and performance changes of BF@Cu after cold-thermal shock cycling; and tables summarizing property changes and performance comparison with other studies (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Dongzhen Chen – School of Materials Science & Engineering, Key Laboratory of Functional Textile Material and Product of the Ministry of Education, Xi'an Key Laboratory of Textile Composites, Xi'an Polytechnic University, Xi'an, Shaanxi 710048, China; orcid.org/0000-0002-6364-6613; Email: chendz365@xpu.edu.cn

Authors

Che Zhou – School of Materials Science & Engineering, Key Laboratory of Functional Textile Material and Product of the Ministry of Education, Xi'an Key Laboratory of Textile Composites, Xi'an Polytechnic University, Xi'an, Shaanxi 710048, China

Lin Zhao – School of Materials Science & Engineering, Key Laboratory of Functional Textile Material and Product of the Ministry of Education, Xi'an Key Laboratory of Textile Composites, Xi'an Polytechnic University, Xi'an, Shaanxi 710048, China

Lei Hui – School of Materials Science & Engineering, Key Laboratory of Functional Textile Material and Product of the

Ministry of Education, Xi'an Key Laboratory of Textile Composites, Xi'an Polytechnic University, Xi'an, Shaanxi 710048, China

Zeyu Yang – School of Materials Science & Engineering, Key Laboratory of Functional Textile Material and Product of the Ministry of Education, Xi'an Key Laboratory of Textile Composites, Xi'an Polytechnic University, Xi'an, Shaanxi 710048, China

Tao Fu – Key Laboratory of Biomedical Information Engineering of Ministry of Education, School of Life Science and Technology, Xi'an Jiaotong University, Xi'an, Shaanxi 710049, China; orcid.org/0000-0002-6356-1871

Jian Huang – College of Chemistry and Chemical Engineering, Xi'an Shiyou University, Xi'an, Shaanxi 710065, China; orcid.org/0000-0001-5942-9432

Complete contact information is available at:

<https://pubs.acs.org/doi/10.1021/acsami.6c07813>

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by the "Textile Vision" Applied Basic Research of China National Textile and Apparel Council (No. J202303), the Natural Science Basic Research Program of Shaanxi (No. 2025JC-YBMS-906), the Science and Technology Planning Project of Xi'an Beilin District (No. GX2401), and the Key Laboratory of Functional Textile Material and Product (Xi'an Polytechnic University), Ministry of Education (No. 2026FTMP006).

■ REFERENCES

- (1) Chen, R.; Wang, X.; Yuan, S.; Chen, H.; Gao, F.; Kong, L. A solution of lunar manufacturing and launch base. *Acta Astronaut.* **2025**, 236, 897–903.
- (2) Neal, C. R.; Salmeri, A.; Abbud-Madrid, A.; Carpenter, J. D.; Colaprete, A.; Hibbitts, K. A.; Kleinhenz, J.; Link, M.; Sanders, G. The Moon needs an international lunar resource prospecting campaign. *Acta Astronaut.* **2024**, 214, 737–747.
- (3) Luo, P.; Zhang, X.; Fu, S.; Li, Y.; Li, C.; Cao, J. First measurements of low-energy cosmic rays on the surface of the lunar farside from Chang'E-4 mission. *Sci. Adv.* **2022**, 8 (2), No. eabk1760.
- (4) Williams, J. P.; Paige, D. A.; Greenhagen, B. T.; Sefton-Nash, E. The global surface temperatures of the Moon as measured by the Diviner Lunar Radiometer Experiment. *Icarus* **2017**, 283, 300–325.
- (5) Yang, W.; Dong, J.; Long, H.; Zhan, P.; Liu, H.; Liu, C.; Shen, C. Flexible polyimide-based conductive composite film with confined carbon nanotubes networks for EMI shielding and joule heating. *Adv. Nanocompos.* **2025**, 2, 322–330.
- (6) Yang, W.; Pan, D.; Liu, S.; Jia, G.; Wang, Y.; Liu, H.; Liu, C.; Shen, C. Multifunctional Wearable Conductive Nanofiber Membrane with Antibacterial and Breathable Ability for Superior Sensing, Electromagnetic Interference Shielding, and Thermal Management. *Adv. Funct. Mater.* **2025**, 35 (6), No. 2414811.
- (7) Wang, Z.; Xing, D.; Yin, R.; Zhan, P.; Liu, C.; Shen, C.; Liu, H. Breathable and waterproof conductive cotton fabric pressure sensor with distinguished electrothermal and electromagnetic interference shielding performances. *Appl. Mater. Today* **2024**, 38, No. 102256.
- (8) Zhang, D.; Yin, R.; Zheng, Y.; Li, Q.; Liu, H.; Liu, C.; Shen, C. Multifunctional MXene/CNTs based flexible electronic textile with excellent strain sensing, electromagnetic interference shielding and Joule heating performances. *Chem. Eng. J.* **2022**, 438, No. 135587.
- (9) Zhang, F.; Long, H.; Yang, W.; Lu, Z.; Liu, H.; Liu, C.; Shen, C. Gradient structural foam-coated fabric for ultra-low reflection

electromagnetic interference shielding. *Composites, Part A* **2026**, *201*, No. 109414.

(10) Naser, M. Z. Extraterrestrial construction materials. *Prog. Mater. Sci.* **2019**, *105*, No. 100577.

(11) Naser, M. Z. Space-native construction materials for earth-independent and sustainable infrastructure. *Acta Astronaut.* **2019**, *155*, 264–273.

(12) Shichalin, O.; Belov, A.; Buravlev, I.; Kolodeznikov, E.; Fedorets, A.; Lembikov, A.; Zolotnikov, S.; Maiorov, V.; Nozdrachev, E.; Ruslan, A.; Azon, S.; Chashchin, A.; Evgeniy, P. Additive manufacturing development of construction materials for a lunar base via spark plasma sintering of volcanic rocks using in-situ resource utilization concept. *Constr. Build. Mater.* **2024**, *442*, No. 137553.

(13) Burke, J. D. Lunar resources: Evaluation and development for use in a lunar base program. *Acta Astronaut.* **1992**, *26* (1), 11–13.

(14) Zeng, L.; Geng, Y.; Li, T.; Zhang, H.; Zhang, R.; Yuan, Z.; Liu, C.; Yang, C.; Li, H.; Li, Y. Effect of ZrO₂-TiO₂ co-doping on the structure, alkali resistance and mechanical properties of basalt fibers. *Constr. Build. Mater.* **2024**, *438*, No. 137312.

(15) Cao, H.; Wang, C.; Chen, J.; Che, X.; Fu, X.; Shi, Y.; Liu, D.; Ling, Z.; Qiao, L.; Lu, X.; Qi, X.; Yin, C.; Liu, P.; Liu, C.; Xin, Y.; Liu, J. A Raman Spectroscopic and Microimage Analysis Perspective of the Chang'e-5 Lunar Samples. *Geophys. Res. Lett.* **2022**, *49* (13), No. e2022GL099282.

(16) Wang, Z.; Li, Y.; Li, J.; Zong, K.; She, Z.; He, Q.; Zhao, J.; Zhang, W.; Zheng, J.; Pan, F.; Luo, T.; Crnobraja, K.; Chen, X.; Xiao, L.; Hu, Z.; Wu, X.; Liu, Y.; Moynier, F. Chemical compositions of Chang'e-6 lunar soil and substantial addition of noritic crust ejecta from Apollo basin. *Geology* **2025**, *53* (7), 557–561.

(17) Wang, R.; Qiao, G. Fibers drawn out from melting lunar regolith simulant: An appealing approach to support the lunar in-situ resource utilization. *Adv. Space Res.* **2025**, *76* (3), 1576–1592.

(18) Ding, M.; Li, Y.; Han, H.; Song, L.; Luo, L.; Li, H.; Wang, Q. Effects of TiO₂ on fiber spinnability, structure, and mechanical properties of glasses derived from simulated lunar soils. *Ceram. Int.* **2024**, *50* (12), 21154–21164.

(19) Guo, Z.-S.; Xing, D.; Xi, X.-Y.; Yue, X.; Liang, C.-G.; Hao, B.; Zheng, Q.; Gutnikov, S. I.; Lazoryak, B. I.; Ma, P.-C. Production of Fibres from Lunar Soil: Feasibility, Applicability and Future Perspectives. *Adv. Fiber Mater.* **2022**, *4* (5), 923–937.

(20) Cao, J.; Peng, H.; Zhang, B.; Yang, Z.; Pan, H.; Chen, J.; Zhao, X. Basalt fiber-reinforced seawater sea-sand geopolymer composites for offshore engineering constructions: Mechanical performance, microstructure, enhanced mechanisms and sustainability. *Constr. Build. Mater.* **2025**, *493*, No. 143129.

(21) Song, S.; Wang, Q.; Ji, D.; Li, L.; Tan, J.; Wu, Q.; Lyu, Y.; Zhang, M. Nacre-Inspired Aramid Nanofibers/Basalt Fibers Composite Paper with Excellent Flame Retardance and Thermal Stability by Constructing an Organic–Inorganic Fiber Alternating Layered Structure. *ACS Appl. Mater. Interfaces* **2024**, *16* (3), 4045–4055.

(22) Liu, J.; Luo, L.; Xu, J.; Zhu, X.; Shi, G.; Wang, Q. The Impact of the Composition on the Properties of Simulated Lunar Mare Basalt Fibers. *Materials* **2024**, *17* (9), 2043.

(23) Li, Y.; Pan, P.; Miao, S.; Feng, Y. Basalt fiber reinforcement mechanism for geopolymer exposed to lunar temperature environment. *Constr. Build. Mater.* **2024**, *451*, No. 138845.

(24) Zhou, S.; Dong, J.; Li, X.; Zhao, X.; Zhang, Q. Continuous Surface Metallization of Polyimide Fibers for Textile-Substrate Electromagnetic Shielding Applications. *Adv. Fiber Mater.* **2023**, *5* (6), 1892–1904.

(25) Tian, J.; Hou, S.; Mao, L.; Xu, X.; Cao, H.; Duan, X.; Li, L.; Zhou, Z.; Ji, Y.; Xie, J.-Q. Printable and filament-drawable PDMS based adhesive assisted manufacturing of highly conductive copper micro-patterns. *J. Colloid Interface Sci.* **2025**, *677*, 130–139.

(26) Chen, H.; Hsu, Y.-C.; Zhai, E.; Bahrami, R.; Lee, M.-H.; Hwang, L.-T.; Lin, Y. C.; Sue, H.-J. Copper-coated carbon fiber composites for electromagnetic interference shielding applications. *Appl. Surf. Sci.* **2025**, *702*, No. 163392.

(27) Zhou, C.; Yu, D.; Xing, C.; Zhu, Z.; Yang, Z.; He, X.; Fu, T.; Song, Z.; Fan, W.; Chen, D. Multi-level metalized PBO composite fabric for polar extreme environments: integrating electromagnetic protection, stable electric thermal conversion, and high-strength. *Chem. Eng. J.* **2025**, *521*, No. 166598.

(28) Azimpour-Shishevan, F.; Akbulut, H.; Mohtadi-Bonab, M. A. Effect of thermal cycling on mechanical and thermal properties of basalt fibre-reinforced epoxy composites. *Bull. Mater. Sci.* **2020**, *43* (1), No. 88.

(29) Fujimori, Y.; Shimizu, M.; Kurashina, T.; Arai, S. Substrate thermal expansion coefficient effect on cracks induced by the high-heat treatment of electroplated Ni-P films for power devices. *Mater. Lett.* **2023**, *350*, No. 134869.

(30) Baytak, T.; Bulut, O. Thermal Stress in Functionally Graded Plates with a Gradation of the Coefficient of Thermal Expansion Only. *Exp. Mech.* **2022**, *62* (4), 655–666.

(31) Zhang, C.; Shen, Q.; Zhang, J.; Luo, G.; Liu, Y.; Sun, Y.; Li, Y. Improved parallelism of graded W–Cu–SiC materials by adjusting the coefficient of thermal expansion. *Ceram. Int.* **2020**, *46* (7), 9714–9721.

(32) Zhang, Q.; Tang, X.; Zhao, Q.; Chen, X.; Wang, K.; Zhang, Q.; Fu, Q. Millefeuille-inspired biomass alternate multilayer composite, for excellent absorption-dominated, broadband EMI shielding and joule heating. *Composites, Part B* **2024**, *283*, No. 111620.

(33) Zhang, B.; Guo, J.; Qu, Z.; Liebscher, M.; Mechtcherine, V.; Kong, Y.; Geng, G.; Wang, F. Toward accurate modeling and assessment of cement-based electromagnetic wave absorbers: Enabling absorption-dominant shielding via multilayered carbon fiber architectures. *Constr. Build. Mater.* **2026**, *506*, No. 144880.

(34) Molla, H. R.; Modarress, H.; Abdouss, M. Electroless nickel–phosphorus deposition on carbon steel CK-75 and study of the effects of some parameters on properties of the deposits. *J. Coat. Technol. Res.* **2012**, *9* (2), 183–188.

(35) Iorio, M.; Santarelli, M. L.; González-Gaitano, G.; González-Benito, J. Surface modification and characterization of basalt fibers as potential reinforcement of concretes. *Appl. Surf. Sci.* **2018**, *427*, 1248–1256.

(36) Mandal, S.; Raj, R.; Samanta, K.; Kumar, S.; Bose, S. Self-Healable Interfaces with Improved Mechanical Properties Induced by Dynamic Network Reconfiguration in Carbon Fiber-Reinforced Epoxy Laminates. *ACS Appl. Mater. Interfaces* **2024**, *16* (25), 31877–31894.

(37) Jiao, Y.; Li, J.; Du, F.; Xu, C.; Zhang, L.; Li, Q. A study of the wear resistance of carbon fiber composites deposited using an alkaline-based palladium-free electroless plating method. *Mater. Today Commun.* **2023**, *37*, No. 106931.

(38) Meng, Y.; Liu, J.; Xia, Y.; Liang, W.; Ran, Q.; Xie, Z. Preparation and characterization of continuous basalt fibre with high tensile strength. *Ceram. Int.* **2021**, *47* (9), 12410–12415.

(39) Dong, B. K.; Wei, C. L.; Lin, J. C.; Xie, L. L.; Liu, K. K.; Xiong, T. J.; Song, W. H.; Tong, P.; Sun, Y. P. A biomimetic aluminum composite exhibiting gradient-distributed thermal expansion, high thermal conductivity, and highly directional toughness. *J. Mater. Sci. Technol.* **2025**, *213*, 90–97.

(40) Xing, D.; Xi, X.-Y.; Ma, P.-C. Factors governing the tensile strength of basalt fibre. *Composites, Part A* **2019**, *119*, 127–133.

(41) Peng, J.; Zhang, R.-Y.; Jia, S.; Liu, W.; Wang, S. Topological near fields generated by topological structures. *Sci. Adv.* **2022**, *8* (41), No. eabq0910.

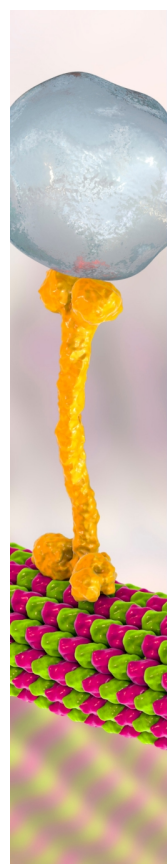
(42) Dao, T. D.; Ishii, S.; Doan, A. T.; Wada, Y.; Ohi, A.; Nabatame, T.; Nagao, T. An On-Chip Quad-Wavelength Pyroelectric Sensor for Spectroscopic Infrared Sensing. *Adv. Sci.* **2019**, *6* (20), No. 1900579.

(43) Wan, J.; Sun, D.; Li, P.; Huang, J.; Chen, Z. Design and Analysis of a Textured Cu-Encapsulated Ni Tube for Low-Reflection Electromagnetic Interference Shielding Material. *Langmuir* **2024**, *40* (17), 9244–9254.

(44) Wang, C.; Hu, C.; Ding, Y.; Li, Z.; Wang, Z.; Lin, X.; Zhou, Y.; Xu, J. h-CoFe₂O₄/Ti₃C₂Tx/BNC Hybrid Aerogels with Modulation Impedance Matching for Electromagnetic Wave Absorption and

- Health Monitoring. *ACS Appl. Mater. Interfaces* **2025**, *17* (1), 2048–2062.
- (45) Lee, S.; Nguyen, N. K.; Kim, M.; Park, P.; Nah, J. Conductivity-Controlled Polyvinylidene Fluoride Nanofiber Stack for Absorption-Dominant Electromagnetic Interference Shielding Materials. *ACS Appl. Mater. Interfaces* **2023**, *15* (27), 33180–33189.
- (46) Luo, C.-L.; Huang, M.-L.; Sun, C.; Zhao, K.-Y.; Hu, Z.; Wang, M. Achieving high joule heating and self-cleaning performance in copper-coated fabrics with excellent microwave shielding. *Cellulose* **2023**, *30* (9), 5987–6000.
- (47) Gao, Y.-N.; Wang, Y.; Yue, T.-N.; Weng, Y.-X.; Wang, M. Multifunctional cotton non-woven fabrics coated with silver nanoparticles and polymers for antibacterial, superhydrophobic and high performance microwave shielding. *J. Colloid Interface Sci.* **2021**, *582*, 112–123.
- (48) Tian, W.; Wang, M.; Liu, Y.; Wang, W. Ohmic heating curing of high content fly ash blended cement-based composites towards sustainable green construction materials used in severe cold region. *J. Cleaner Prod.* **2020**, *276*, No. 123300.
- (49) Lin, K.; Pang, S. D. The influence of thermal residual stresses and thermal generated dislocation on the mechanical response of particulate-reinforced metal matrix nanocomposites. *Composites, Part B* **2015**, *83*, 105–116.
- (50) Zhao, J.; Zhang, R.; Guo, Z.; Zhu, Q.; Li, Q. Synergistic gradient CTE-Metallurgical bonding for high-performance Cu/graphite composites. *Appl. Surf. Sci.* **2026**, *724*, No. 165710.
- (51) Ding, Q.; Yang, J.; Gu, S.; Chen, C.; Cheng, Z.; Liu, Y.; Yan, P.; Si, M.; Lu, S.; Zhang, Z.; Wang, L.; Fan, Y.; Jiang, W. Novel fire-resistant SiBCN fiber paper with efficient electromagnetic interference shielding and Joule-heating performance. *Chem. Eng. J.* **2024**, *497*, No. 154485.
- (52) Li, W.; Song, Z.; He, Y.; Zhang, J.; Bao, Y.; Wang, W.; Sun, Z.; Ma, Y.; Liu, Z.; Niu, L. Natural sedimentation-assisted fabrication of Janus functional films for versatile applications in Joule heating, electromagnetic interference shielding and triboelectric nanogenerator. *Chem. Eng. J.* **2023**, *455*, No. 140606.
- (53) Hwang, U.; Cho, S.; Koo, C. M.; Nam, J.-D.; Sung, J. Nanoporous Conductive Carbon Films via High-Carbon-Yield Carbonization of Poly(p-phenylene benzobisoxazole) Nanofibers. *Small* **2026**, *22* (13), No. e14183.
- (54) Yin, G.; Wang, Y.; Wang, W.; Yu, D. Multilayer structured PANI/MXene/CF fabric for electromagnetic interference shielding constructed by layer-by-layer strategy. *Colloids Surf., A* **2020**, *601*, No. 125047.
- (55) Yan, J.; Chen, M.; Tan, R.; Lin, C.; Jiang, S.; Wang, W.; Pan, S.; Xiao, H.; Ren, E.; Guo, R. Flexible multifunctional MXene@Ag nanowires/cotton fabric inspired by transport of nutrients by roots for electromagnetic shielding, infrared stealth, joule/solar heating and flame retardancy. *J. Mater. Chem. A* **2024**, *12* (47), 33162–33176.
- (56) Song, S.; Li, L.; Ji, D.; Zhao, J.; Wu, Q.; Wang, Q. Flexible Basalt Fiber/Aramid Nanofiber/Carbon Nanotube Electromagnetic Shielding Paper with Outstanding Environmental Stability and Joule Heating Performance. *ACS Appl. Mater. Interfaces* **2023**, *15* (29), 35495–35506.
- (57) Singh, K.; Baheti, V. EMI shielding and joule heating applications of woven and nonwoven activated carbon fabrics of Kevlar fibres. *J. Anal. Appl. Pyrolysis* **2025**, *188*, No. 107041.
- (58) Singh, K.; Baheti, V. Electromagnetic interference shielding and ohmic heating of layered structures of activated carbon fabric and metalized fabric. *Synth. Met.* **2023**, *298*, No. 117439.
- (59) Gong, K.; Yin, L.; Qian, X.; Wu, Z.; Liu, H.; Zhang, J.; Zhou, K.; Yu, W. W. Dual-Doped Polyaniline Boosting the Construction of Fireproof MXene/Aramid Nanofiber Films for Superior Electromagnetic Interference Shielding and On-Demand Joule Heating. *ACS Appl. Polym. Mater.* **2026**, *8* (3), 1687–1697.
- (60) Wang, C.; Tian, L.; Li, J.; Wu, H.; Li, S.; Ma, J.; Wang, Y.; Fan, W.; Ni, J.; Hou, F.; Wang, Y.; Zhu, C.; Li, T. Monolithic Self-Supported MXene/Aramid Nanofiber/Ag Nanoparticle/MXene

Composite Membranes for Integrated EMI Shielding, Joule Heating, and Energy Storage. *ACS Appl. Nano Mater.* **2026**, *9* (5), 2364–2376.



CAS BIOFINDER DISCOVERY PLATFORM™

**PRECISION DATA
FOR FASTER
DRUG
DISCOVERY**

CAS BioFinder helps you identify targets, biomarkers, and pathways

Unlock insights

CAS
A division of the
American Chemical Society